



Course Title: Data Structure & Algorithm Lab II

Course Code: CSE 2218

Trimester & Year: Spring 2026

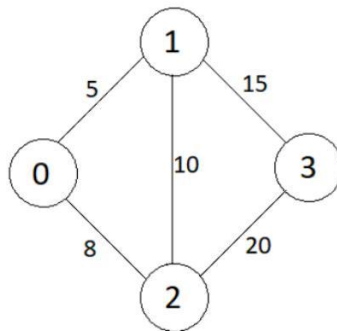
Section: F

Credit Hours: 1.0

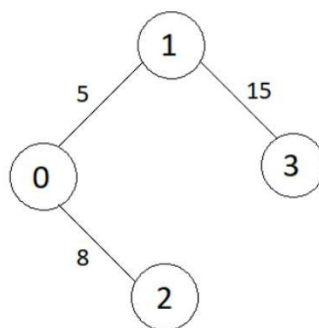
Q1: Prim's MST

You are given an undirected connected weighted graph having '**N**' nodes numbered from '**1**' to '**N**'. A matrix '**E**' of size **M x 2** is given which represents the '**M**' edges such that there is an edge directed from node **E[i][0]** to node **E[i][1]**. You are supposed to return the minimum spanning tree where you need to return weight for each edge in the MST.

For example:



The MST (Minimum Spanning Tree) for the above graph is:



Input Format:

The first line Contains an integer '**T**' representing the number of the test case. Then the test cases are as follows.

The first line of each test case argument contains a given integer '**N**' representing the number of nodes in the graph.



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The second line of each test case contains a given integer '**M**' representing the number of edges in the graph.

The next '**M**' lines in each test case contain a matrix '**E**' of size $M \times 2$ which represents the '**M**' edges such that there is an edge directed from node $E[i][0]$ to node $E[i][1]$.

Output Format:

For each test case, print the minimum spanning tree in the form of edges and their weights which are included in the MST.

Sample Input 1:

```
1
5 14
1 2 2
1 4 6
2 1 2
2 3 3
2 4 8
2 5 5
3 2 3
3 5 7
4 1 6
4 2 8
4 5 9
5 2 5
5 3 7
5 4 9
```

Sample Output 1:

```
1 2 2
1 4 6
2 3 3
2 5 5
```

Explanation of Input 1 :

The Minimum spanning tree for the given graph will contain the edges: (1,2) with weight 2, (1,4) with weight 6, (2,3) with weight 3 and (2,5) with weight 5.

Sample Input 2 :

```
1
5 15
1 2 21
1 4 16
```



2 1 12

2 3 13

2 4 18

2 5 15

3 2 13

3 5 17

4 1 16

4 2 18

4 5 19

5 1 18

5 2 15

5 3 17

5 4 19

Sample Output 2 :

1 2 12

1 4 16

2 3 13

2 5 15

Explanation of Input 2 :

The Minimum spanning tree for the given graph will contain the edges: (1,2) with weight 12, (1,4) with weight 16, (2,3) with weight 13 and (2,5) with weight 15.

Tip:

Create a function like the following that will find the MST and return it:

```
vector<pair<pair<int, int>, int>> calculatePrimsMST(int n, int m, vector<pair<pair<int, int>, int>> &g)
{
    // Write your code here.
}
```



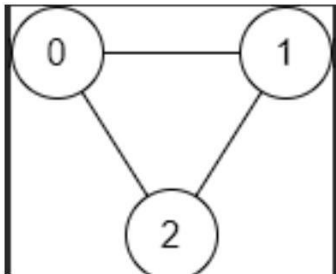
Q2: Find if Path Exist in Graph

There is a bi-directional graph with **n vertices**, where each vertex is labeled from **0 to n - 1** (inclusive). The edges in the graph are represented as a 2D integer array **edge**, where each **edges[i] = [u_i, v_i]** denotes a bi-directional edge between vertex, u_i and vertex, v_i. Every vertex pair is connected by at most one edge, and no vertex has an edge to itself.

You want to determine if there is a valid path that exists from vertex source to vertex destination.

Given edges and the integers n, source, and destination, return true if there is a valid path from source to destination, or false otherwise.

Example 1:



Input: n = 3, edges = [[0,1],[1,2],[2,0]], source = 0, destination = 2

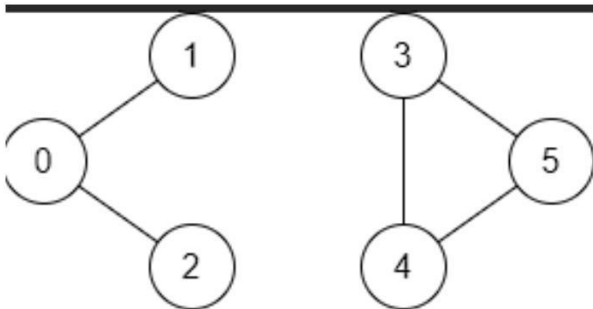
Output: true

Explanation: There are two paths from vertex 0 to vertex 2:

- 0 → 1 → 2

- 0 → 2

Example 2:



Input: n = 6, edges = [[0,1],[0,2],[3,5],[5,4],[4,3]], source = 0, destination = 5

Output: false

Explanation: There is no path from vertex 0 to vertex 5.

Constraints:

$1 \leq n \leq 2 * 10^5$

$0 \leq \text{edges.length} \leq 2 * 10^5$

$\text{edges}[i].\text{length} == 2$

$0 \leq u_i, v_i \leq n - 1$



$u_i \neq v_i$

$0 \leq \text{source}, \text{destination} \leq n - 1$

There are no duplicate edges.

There are no self edges

Q3: Home Delivery

Mr. Chandler has started Home Delivery scheme in one of his restaurants. As the scheme is new, he appoints only one employee to deliver food to various locations. However, He is facing trouble with that delivery boy as he is an absent-minded chap!

The delivery boy always forgets to fill fuel in his delivery scooter. So, what he does is that whenever Mr. Chandler sends him for delivery, he goes to the gas station from the restaurant first. He gets his tank filled and then he heads towards his destination. He will do this every single time *irrespective of the destination*. The delivery boy tries his best to be on time. And to do this, he will choose those paths **(from restaurant to gas station AND gas station to destination)** which cost him the **least amount of time**.

Your task is to tell Mr. Chandler how much time the delivery boy can save if he had enough fuel in his scooter i.e. if he went to the destination directly without stopping for fuel (taking the path which costs him least amount of time). The city has **N** streets numbered from **0** to **N-1**. The restaurant is on street number **S**, the gas station is on street number **G** and the food has to be delivered to street **D**. Note that **S**, **G** and **D** need **not** be distinct.

Input:

- First line of the input contains a single integer **N**.
- Then follows an **N x N** matrix **T** which is represented in **N** lines with **N** space separated integers on each line. **T[i][j]** denotes the time taken to move from the i^{th} street to j^{th} street. Obviously, **T[i][i] = 0**.
- Next line contains a single integer **M**, the number of scenarios. The following **M** lines contain 3 space separated integers **S**, **G** and **D**.

Output:

For each of the **M** scenarios, output the time taken by the delivery boy to deliver the food and the time he could have saved if he went directly from **S** to **D**. Both these values must be on the same line separated by a single space.



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Sample Input	Sample Output	Explanation
4 0 2 1 3 1 0 4 5 3 1 0 3 1 1 1 0 4 0 2 1 0 2 2 3 1 2 3 0 1	2 0 1 0 3 2 3 2	Look, in the first scenario: The Delivery Boy has travelled 0 (S) to 2(G) and then 2 (G) to 1 (D). ➤ Least time from 0->2 = 1 ➤ Least time from 2->1 = 1 ➤ Total Time taken is 2. If He would have the fuel and travelled from 0 to 1 directly: ➤ Least time from 0->1 = 2 Thus, He could have saved (2-2) = 0 That's why first line of output 2 0. The rest follows the same theory.

Q4: The Secret Magic Trip

Marvel is a land with N cities (numbered 1 through N) connected by M bidirectional roads. Dr. Strange is on a mission to spread a secret Magic. He has a sequence $A_1, A_2, \dots, A_K$ and he must visit cities in the following way: he starts in the city A_1 , travels to the city A_2 , then travels from there to the city A_3 and so on until he reaches the city A_K . Note that a city may appear in the sequence A for multiple times, but $A_i \neq A_{i+1}$ for each valid i . When travelling between two cities, Dr. Strange always follows one of the shortest paths between them (but not necessarily the same one if he travels between them again).

The government is trying to track which cities are targeted by Dr. Strange. However, they do not know the sequence A . Instead, they have tracked Dr. Strange's movement as a sequence $B_1, B_2, \dots, B_L$ of cities visited by him during his mission. Formally, $B_1 = A_1$ is the city where Dr. Strange starts, then for each valid i , he moves from the city B_i to the city B_{i+1} using the direct road between them and finally, he ends his mission in the city $B_L = A_K$; Dr. Strange's sequence A is a subsequence of B , since the sequence B also contains all shortest paths which Dr. Strange followed. It is guaranteed that there is a direct road between cities B_i and B_{i+1} for each valid i .

Help the government of Marvel find the minimum possible number of targeted cities, i.e. the minimum possible value of K for a sequence A that corresponds to the given sequence B , or determine that no such sequence A exists.



Input:

The first line of the input contains a single integer T denoting the number of test cases. The description of T test cases follows.

- The first line of each test case contains three space-separated integers N , M and L .
- The second line contains L space-separated integers $B_1, B_2, \dots, B_L$
- Each of the following M lines contain three space-separated integers u , v , and w describing a road between cities u and v with length w .

Output:

For each test case, print a single line containing one integer — the minimum possible K , or -1 if a valid sequence A does not exist.

Input:	Output:
3	6
3 3 6	5
1 2 3 1 2 3	-1
1 2 1	
2 3 1	
3 1 1	
4 4 9	
1 2 3 4 1 2 3 4 1	
1 2 1	
2 3 1	
3 4 1	
4 1 1	
3 3 2	
1 2	
1 2 3	
2 3 1	
3 1 1	